

# PROJECT REPORT

## Auto-Ticket Classification using Flow Designer

ServiceNow School IT Helpdesk Automation

| Team Member      | Primary Responsibility                                |
|------------------|-------------------------------------------------------|
| Aditya A         | Sprint 1 - Requirement Analysis & Planning            |
| Akash N A        | Sprint 2 - Backend Development & Schema Configuration |
| Radha Krishnan P | Sprint 3 - Flow Designer Automation                   |
| Ferdin Joe A     | Sprint 4 - Testing & Validation                       |
| Simeon P         | Sprint 5 - Deployment & Packaging                     |

# TABLE OF CONTENTS

## **1. INTRODUCTION**

1.1 Project Overview

1.2 Purpose

## **2. IDEATION PHASE**

2.1 Problem Statement

2.2 Empathy Map Canvas

2.3 Brainstorming

## **3. REQUIREMENT ANALYSIS**

3.1 Customer Journey Map

3.2 Solution Requirement

3.3 Data Flow Diagram

3.4 Technology Stack

## **4. PROJECT DESIGN**

4.1 Problem Solution Fit

4.2 Proposed Solution

4.3 Solution Architecture

## **5. PROJECT PLANNING & SCHEDULING**

5.1 Project Planning

## **6. FUNCTIONAL AND PERFORMANCE TESTING**

6.1 Performance Testing

## **7. RESULTS**

7.1 Output Screenshots

## **8. ADVANTAGES & DISADVANTAGES**

## **9. CONCLUSION**

## **10. FUTURE SCOPE**

## **11. APPENDIX**

# 1. INTRODUCTION

## 1.1 Project Overview

The Auto-Ticket Classification using Flow Designer project addresses the operational challenge of classifying incoming IT helpdesk requests in a school environment. In many educational settings, students submit support tickets with vague or non-standard descriptions. At the same time, students often lack the technical understanding needed to select appropriate technical categories and subcategories. This puts an unnecessary triage burden on helpdesk technicians, who must review, reclassify, and route tickets before troubleshooting can begin.

The implemented solution leverages ServiceNow Flow Designer to evaluate keywords in ticket short descriptions automatically upon creation. When supported issue keywords are detected, the system populates Category and Subcategory fields with consistent choices and sends an immediate email acknowledgment to the caller. This reduces manual effort, speeds up processing, and provides structured records for reporting.

## 1.2 Purpose

The purpose of this project is to:

- Simplify the ticket submission experience for students by removing the need for technical category selection.
- Automate the assignment of Category and dependent Subcategory fields based on issue descriptions.
- Send automatic acknowledgment notifications to the ticket requester.
- Provide a structured custom table (u\_school\_it\_ticket) with controlled access and change tracking via ServiceNow Update Sets.

# 2. IDEATION PHASE

## 2.1 Problem Statement

The project defines two core problem statements: for the student requester and for the helpdesk technician.

**Student Problem Statement:** Students trying to submit an IT support request quickly and correctly face confusion because they do not know which technical category or subcategory to select, causing hesitation and delayed reporting.

**Technician Problem Statement:** Helpdesk technicians trying to process and resolve tickets efficiently must manually inspect and reclassify incoming requests because students frequently pick incorrect categories or leave them empty, leading to unnecessary triage delays.

## 2.2 Empathy Map Canvas

The empathy mapping process confirmed that students think primarily about getting back to classwork, see complex category forms as intimidating, say "Wi-Fi is not working", and desire simple ticket forms with automated confirmation. Technicians think about meeting resolution targets, see disarranged queues, and gain immense triage time savings when categorization happens automatically.

## 2.3 Brainstorming

The team brainstormed and prioritized multiple approaches. A rule- and keyword-based ServiceNow Flow Designer implementation was selected over complex machine-learning models due to its deterministic accuracy, platform-native execution, rapid response time, and zero licensing overhead.

### 3. REQUIREMENT ANALYSIS

#### 3.1 Customer Journey Map

The customer journey reflects the student's progression from encountering an IT glitch through to resolution. The student enters the portal, submits a ticket with Caller and Short Description, Flow Designer evaluates the issue, and an email acknowledgment is delivered instantly.

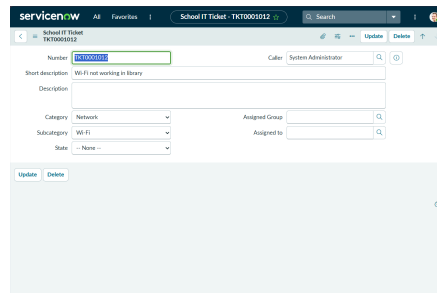
The image shows a screenshot of a web browser displaying the 'servicenow' portal. The browser's address bar shows 'School IT Ticket: TK1000012'. The page title is 'School IT Ticket: TK1000012'. The form contains the following fields: 'Number' (TK1000012), 'Caller' (System Administrator), 'Short description' (Wi-Fi not working in library), 'Description' (empty), 'Category' (Network), 'Subcategory' (Wi-Fi), 'State' (None), 'Assigned Group' (empty), and 'Assigned to' (empty). At the bottom of the form are 'Update' and 'Delete' buttons.

Figure 1: School IT Ticket form used during the ticket-submission journey.

#### 3.2 Solution Requirement

Functional requirements specify ticket creation (FR-1), keyword-based categorization (FR-2), dependent choices (FR-3), outbound notification (FR-4), queue review (FR-5), and Update Set change tracking (FR-6).

Non-functional requirements guarantee simple form usability (NFR-1), role-based ACL security (NFR-2), schema consistency (NFR-3), sub-second execution performance (NFR-4), and cloud platform availability (NFR-5).

### 3.3 Data Flow Diagram

The data flow begins with the student entering ticket information. The custom record is stored in ServiceNow, Flow Designer evaluates the Short Description, and the resulting classification and notification support the helpdesk process.

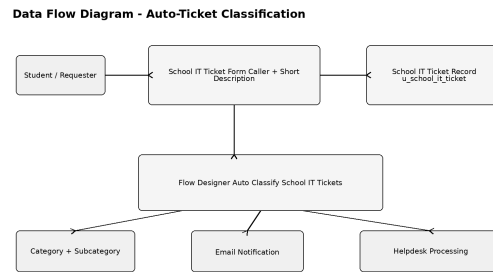


Figure 2: Data Flow Diagram for the automatic ticket-classification process.

#### Data Flow Diagram (Figure 2) Detailed Explanation:

The DFD illustrates the end-to-end information path: (1) The Student enters identity and problem text into the School IT Ticket Form; (2) The record is committed into the custom table (`u_school_it_ticket`); (3) Insertion of an unclassified ticket triggers Flow Designer; (4) The flow evaluates Short Description keywords against configured conditions; (5) Flow Designer updates Category and Subcategory in the record; (6) An outbound email action notifies the requester; (7) The pre-classified ticket is made available in the technician's queue for resolution.

### 3.4 Technology Stack

The solution is implemented entirely on the ServiceNow cloud platform without external application frameworks.

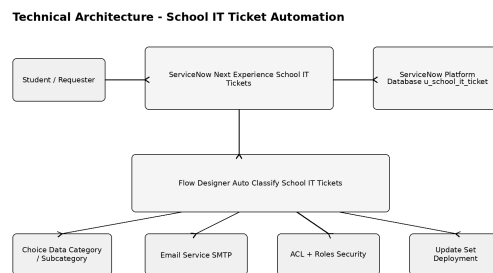


Figure 3: Technical architecture of the ServiceNow implementation.

#### Technical Architecture (Figure 3) Detailed Explanation:

The architecture consists of: (1) User Interface via ServiceNow Next Experience; (2) Custom Data Store in the platform database (`u_school_it_ticket`); (3) Automation Engine powered by ServiceNow Flow Designer; (4) Notification Engine utilizing platform SMTP; (5) Security Layer governed by ACLs and roles; (6) Change Management using ServiceNow Update Sets.

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

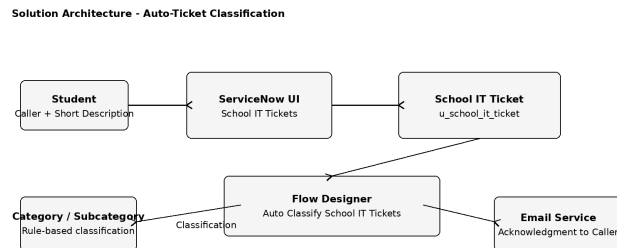
The solution fits the identified problem by shifting supported classification work from the requester to the ServiceNow workflow. The student supplies basic information that is natural to provide, while the workflow applies the internal classification rules.

For the implemented prototype, Wi-Fi or Network language maps to Network / Wi-Fi, projector-related language maps to Hardware / Projector, password or login language maps to Access / Forgot Password, and slow or hanging computer language maps to Performance / Slow Computer.

### 4.2 Proposed Solution

- Create the School IT Ticket custom table and School IT Support application/module.
- Capture Caller and Short Description from the requester.
- Use dependent Category and Subcategory choices.
- Trigger the classification flow when a new ticket is created with an empty Category.
- Apply branches for network, hardware, access and performance keywords.
- Send an acknowledgment email to the Caller.
- Use an Update Set to track and export the configuration for deployment.

### 4.3 Solution Architecture



*Figure 4: Solution architecture connecting the ServiceNow user interface, ticket data, Flow Designer and notification.*

#### **Solution Architecture (Figure 4) Detailed Explanation:**

Figure 4 connects user interactions directly to underlying platform services. Student inputs from the UI populate the custom schema. The Flow Designer execution engine asynchronously processes the record without degrading UI responsiveness, evaluates condition logic, performs field updates, and invokes the notification subsystem.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

| Sprint   | Schedule                  | Member           | Milestone                                  | Points |
|----------|---------------------------|------------------|--------------------------------------------|--------|
| Sprint-1 | 28 Sep 2026 - 28 Sep 2026 | Aditya A         | Requirement Analysis & Planning            | 5      |
| Sprint-2 | 28 Sep 2026 - 28 Sep 2026 | Akash N A        | Backend Development & Schema Configuration | 11     |
| Sprint-3 | 29 Sep 2026 - 29 Sep 2026 | Radha Krishnan P | Flow Designer Automation                   | 11     |
| Sprint-4 | 30 Sep 2026 - 30 Sep 2026 | Ferdin Joe A     | Testing & Validation                       | 5      |
| Sprint-5 | 01 Oct 2026 - 01 Oct 2026 | Simeon P         | Deployment & Packaging                     | 5      |
| Total    | -                         | Project Team     | All five milestones                        | 37     |

The five one-day sprints were assigned to the team members specified for this project. The 37 story-point total is a planning estimate mapped to the documented project work items.

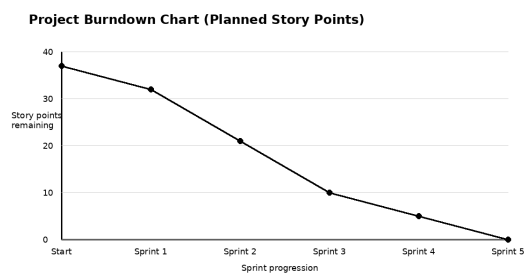


Figure 5: Planned burndown chart based on the 37-point project plan.

#### Burndown Chart (Figure 5) Explanation:

The burndown progression illustrates the execution velocity: beginning with 37 points, Sprint 1 burns down to 32 points, Sprint 2 to 21 points, Sprint 3 to 10 points, Sprint 4 to 5 points, and Sprint 5 closes all remaining points at release, achieving an average team velocity of 7.4 story points per sprint.

## 6. FUNCTIONAL AND PERFORMANCE TESTING

## 6.1 Performance Testing

The supplied testing format requires test scenarios, steps, expected and actual results, and pass/fail status. The implemented project provides functional evidence but does not include measured response-time, concurrent API, or file-upload benchmark results. Those unsupported measures are therefore recorded as Not Tested or N/A.

| Test ID | Scenario                            | Expected Result                                   | Actual Result                                                    | Status     |
|---------|-------------------------------------|---------------------------------------------------|------------------------------------------------------------------|------------|
| FT-01   | Create School IT Ticket             | Record created with Caller and Short Description. | Created School IT Ticket records are visible in the ticket list. | Pass       |
| FT-02   | Automatic Wi-Fi Classification      | Category = Network and Subcategory = Wi-Fi.       | The demonstrated ticket shows Network and Wi-Fi.                 | Pass       |
| FT-03   | Projector / Hardware Classification | Projector request maps to Hardware / Projector.   | A projector request is shown as Hardware in the ticket list.     | Pass       |
| FT-04   | Email Acknowledgment                | Acknowledgment email is generated for the Caller. | Email log and preview show the acknowledgment message.           | Pass       |
| PT-01   | Response Time Measurement           | Execution time is measured and recorded.          | No timing measurement was supplied.                              | Not Tested |
| PT-02   | Concurrent API Load                 | External API remains responsive.                  | N/A; no external classification API is used.                     | N/A        |
| PT-03   | File Upload Load                    | Multiple files process without failure.           | N/A; file upload is not part of the workflow.                    | N/A        |

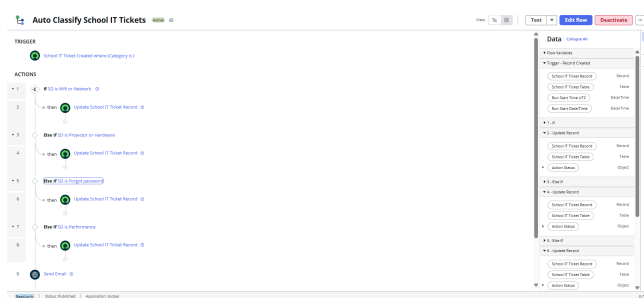


Figure 6: Published Flow Designer workflow used for the functional classification process.

7. RESULTS

7.1 Output Screenshots

The screenshots below document the implemented ServiceNow workflow, classification result, notification, and deployment package.

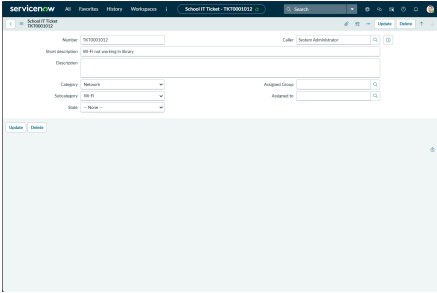


Figure 7: School IT Ticket showing automatic Category and Subcategory classification.

A screenshot of the ServiceNow 'School IT Tickets' list view. It shows a table with the following columns: Number, Assigned Group, Assigned To, Caller, Category, Description, Short description, State, and Subcategory. The table contains four rows of data.

| Number     | Assigned Group | Assigned To | Caller               | Category | Description             | Short description            | State | Subcategory |
|------------|----------------|-------------|----------------------|----------|-------------------------|------------------------------|-------|-------------|
| 7470000003 | Service        | Service     | Service              |          |                         | SR IT not working in library |       |             |
| 7470000002 | Service        | Service     | System Administrator | Hardware | Problems not working in | SR IT not working in library |       |             |
| 7470000001 | Service        | Service     | System Administrator |          |                         | SR IT not working in library |       |             |
| 7470000002 | Service        | Service     | System Administrator | Network  |                         | SR IT not working in library |       |             |

Figure 8: School IT Tickets list showing created records and classification results.



## 7.1 Output Screenshots - Configuration Evidence

The screenshot shows the 'Dictionary Entry - New record' configuration page. The 'Table' is 'School IT Ticket [s\_school\_it\_ticket\_...]' and the 'Type' is 'String'. The 'Column label' is 'Short description' and the 'Column name' is 's\_short\_description'. The 'Max length' is 180. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Function field' checkbox is unchecked. The 'Mandatory' checkbox is unchecked. The 'Display' checkbox is unchecked. The 'Read only option' is set to 'None'. The 'Default value' field is empty. The 'Submit' button is at the bottom right.

Figure 12: Short Description field configuration.

The screenshot shows the 'Dictionary Entry - Caller' configuration page. The 'Table' is 'School IT Ticket [s\_school\_it\_ticket\_...]' and the 'Type' is 'Reference'. The 'Column label' is 'Caller' and the 'Column name' is 's\_caller'. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Function field' checkbox is unchecked. The 'Mandatory' checkbox is unchecked. The 'Display' checkbox is unchecked. The 'Read only option' is set to 'None'. The 'Default value' field is empty. The 'Delete Columns' and 'Update' buttons are at the bottom.

Figure 13: Caller reference field configuration.

The screenshot shows the 'Dictionary Entry - New Record' configuration page. The 'Table' is 'School IT Ticket [s\_school\_it\_ticket\_...]' and the 'Type' is 'Choice'. The 'Column label' is 'State' and the 'Column name' is 's\_state'. The 'Max length' is 180. The 'Application' is 'Global'. The 'Active' checkbox is checked. The 'Function field' checkbox is unchecked. The 'Mandatory' checkbox is unchecked. The 'Display' checkbox is unchecked. The 'Read only option' is set to 'None'. The 'Default value' field is empty. The 'Submit' button is at the bottom right.

Figure 14: Category field configuration.

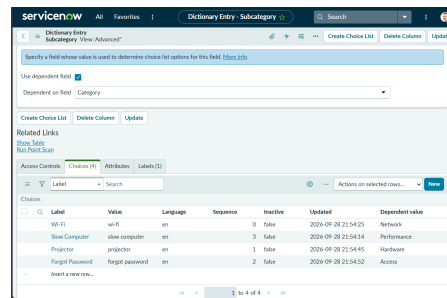


Figure 15: Subcategory configured as dependent on Category.

## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Simple requester experience with basic input fields.
- Automatic classification for supported issue keywords.
- Consistent Category/Subcategory mapping.
- Immediate requester acknowledgment.
- Structured custom ticket storage.
- Controlled change tracking through Update Sets and XML export.

### Disadvantages

- Keyword rules depend on how the problem is written.
- Unsupported or vague descriptions may require manual review.
- The current prototype is rule-based rather than model-based.
- No measured performance benchmark is included.
- Automatic technician assignment is not implemented.

## 9. CONCLUSION

The Auto-Ticket Classification using Flow Designer project demonstrates a complete ServiceNow workflow for simplifying school IT helpdesk ticket processing. The implementation combines a custom data model, dependent choices, event-driven classification, email acknowledgment, access control, and Update Set based deployment tracking.

The supplied screenshots provide functional evidence for ticket creation, automatic classification, notification and finalization. The completed project package can be stored and versioned using the exported Update Set XML.

## 10. FUTURE SCOPE

- Expand keyword coverage and add fallback handling for ambiguous requests.
- Introduce automatic assignment to support groups.
- Add status-change and resolution notifications.
- Add dashboards for ticket volume, classification and response trends.
- Measure response time and perform controlled load testing.
- Evaluate AI-assisted classification as a future enhancement with appropriate testing and governance.

## 11. APPENDIX

### Source Code (if any)

No separate application source code is required for the implemented prototype. The main implementation is ServiceNow configuration: custom table and fields, choices, Flow Designer logic, email notification, ACLs and Update Set configuration. The completed Update Set was exported as an XML deployment package.

### Dataset Link

Not applicable. The implemented workflow does not depend on a machine-learning dataset.

### GitHub & Project Demo Link

**GitHub Repository:** <https://github.com/Akash717656/Auto-Ticket-Classification-using-Flow-Designer>

**Project Tracking / Review Board:** [Naan Mudhalvan SkillWallet Project Portal](#)

**Live public demo:** Not provided. The application runs in the ServiceNow environment used for the project.